SKILLS REASONING (6)

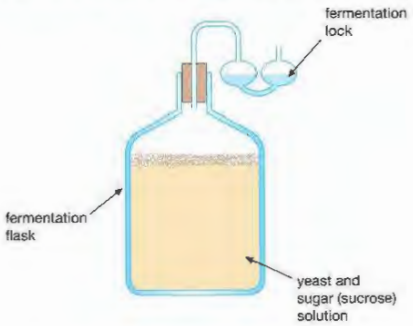
c Explain how a steady temperature is maintained in the fermenter.

d What is supplied through the inlet marked 'X'?

(7) e Suggest what would happen to the growth of *Penicillium* in the fermenter if the paddles stopped working.

SKILLS CRITICAL THINKING (8)

6 The diagram shows a fermentation flask used to make wine at home.



a The yeast cells in the flask respire anaerobically. Write a word equation for this process.

SKILLS REASONING (7)

b Yeast cells can also respire aerobically. How is this prevented in this flask?

c Home winemaking flasks like this are often kept in a warm airing cupboard. Explain why this is done.

SKILLS CRITICAL THINKING (7)

d Wine usually contains no more than about 15% ethanol (alcohol). Explain why.

7 Answer these questions about making yoghurt. Try at first to answer them without looking back to the section on page 285

a Why is the milk pasteurised at the start of the process?

b Why is the mixture of milk and bacteria incubated at 45 °C?

c What causes the milk to thicken?

d Why does fermentation eventually stop?

(8) e Explain how making yoghurt is a way of preserving the nutrients from milk.

22 GENETIC MODIFICATION

In this chapter we will look at ways in which it is now possible to manipulate genes and produce genetically modified organisms – the science of 'genetic engineering'.

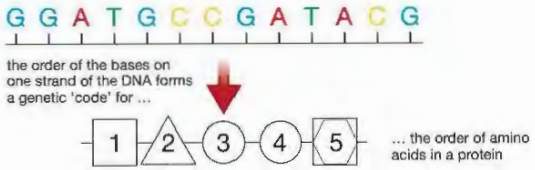
LEARNING OBJECTIVES

- Understand that the term 'transgenic' means the transfer of genetic material from one species to a different species
- Understand how restriction enzymes are used to cut DNA at specific sites and ligase enzymes are used to join pieces of DNA together
- Understand how plasmids and viruses can act as vectors, which take up pieces of DNA, and then insert this recombinant DNA into other cells
- Understand how large amounts of human insulin can be manufactured from genetically modified bacteria that are grown in a fermenter
- Understand how genetically modified plants can be used to improve food production

BIOLOGY ONLY

- Understand how cloned transgenic animals can be used to produce human proteins.

In Chapter 16 you saw that a gene is a section of a molecule of DNA that codes for the production of a protein. The coding strand of the DNA contains triplets of bases, each triplet coding for one amino acid. Different genes produce different proteins because each has a unique sequence of bases that codes for a unique sequence of amino acids (Figure 22.1).



▲ Figure 22.1 The role of DNA in protein synthesis

The protein that is produced could be:

- an enzyme that controls a particular reaction inside a cell or in the digestive system
- a structural protein like keratin in hair, collagen in skin or one of the many proteins found in the membranes of cells
- a protein hormone such as insulin
- a protein with a specific function such as haemoglobin or an antibody.

KEY POINT

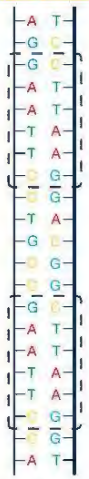
A transgenic organism is one that contains a gene that has been introduced from another species. For example, some bacteria have had human genes transferred to them that allow them to make human insulin. Some sheep secrete the protein alpha-1-antitrypsin (AAT) in their milk because they have the human gene that directs the manufacture of this substance. Because they contain 'foreign' genes, they are no longer quite the same organisms. They are transgenic.

RECOMBINANT DNA

The production of **recombinant DNA** is the basis of genetic engineering. A section of DNA – a gene – is cut out of the DNA of one species and inserted into the DNA of another. This new DNA is called 'recombinant' DNA because the DNA from two different organisms has been 'recombined'. The organism that receives the gene from a different species is a **transgenic** organism.

The organism receiving the new gene now has an added capability. It will manufacture the protein that the new gene codes for. For example, a bacterium receiving the human gene that codes for insulin production will make human insulin. If these transgenic bacteria are cultured by the billion in a fermenter, they become a factory for making human insulin.

PRODUCING GENETICALLY MODIFIED (TRANSGENIC) BACTERIA



▲ Figure 22.2 Part of a DNA molecule containing the base sequence G-A-A-T-T-C. Notice that the sequence is present on both strands, but running in opposite directions.

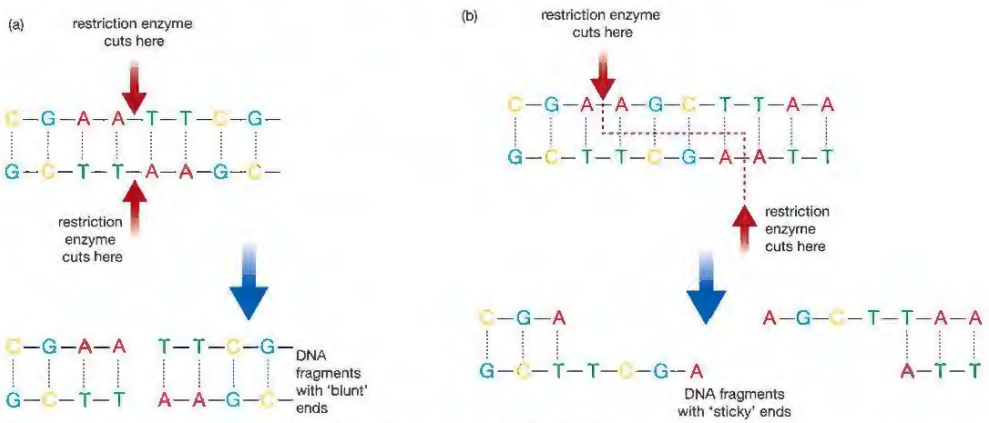
The breakthrough in being able to transfer DNA from cell to cell came when it was found that bacteria have two sorts of DNA – the DNA found in their bacterial 'chromosome' and much smaller circular pieces of DNA called **plasmids** (see Unit 1, Figure 2.10).

Bacteria naturally 'swap' plasmids, and biologists found ways of transferring plasmids from one bacterium to another. The next stage was to find molecular 'scissors' and a molecular 'glue' that could cut out genes from one molecule of DNA and stick them back into another. Further research found the following enzymes that were able to do this.

- **Restriction endonucleases** (usually shortened to **restriction enzymes**) are enzymes that cut DNA molecules at specific points. Different restriction enzymes cut DNA at different places. They can be used to cut out specific genes from a molecule of DNA.
- **Ligases** (or DNA ligases) are enzymes that join the cut ends of DNA molecules.

Each restriction enzyme recognises a certain base sequence in a DNA strand. Wherever it encounters that sequence, it will cut the DNA molecule. Suppose a restriction enzyme recognises the base sequence G-A-A-T-T-C. It will only cut the DNA molecule if it can 'see' the base sequence on both strands. Figure 22.3 illustrates this.

Some restriction enzymes make a straight cut and the fragments of DNA they produce are said to have 'blunt ends' (Figure 22.3(a)). Other restriction enzymes make a staggered cut. These produce fragments of DNA with overlapping ends with complementary bases. These overlapping ends are called 'sticky ends' because fragments of DNA with exposed bases are more easily joined by ligase enzymes (Figure 22.3(b)).

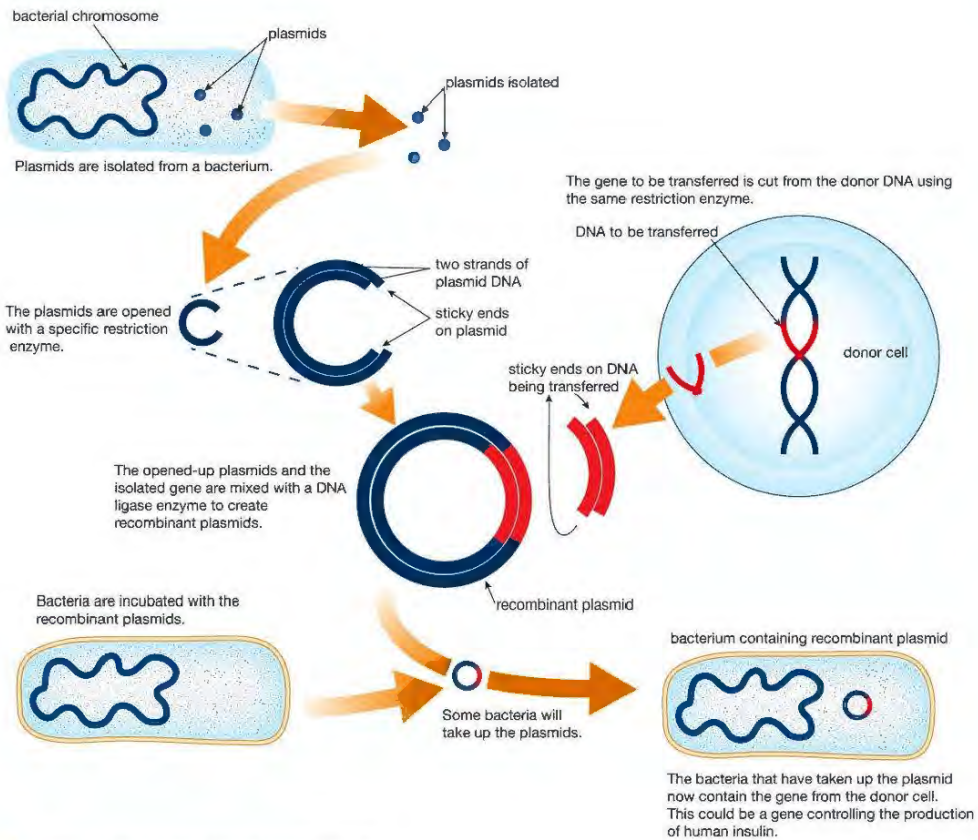


▲ Figure 22.3 How restriction enzymes cut DNA. (a) Forming blunt ends. (b) Forming sticky ends.

EXTENSION WORK

There is a lot more to producing recombinant DNA and transgenic bacteria than is shown here. You could carry out an some research to find out more about this topic.

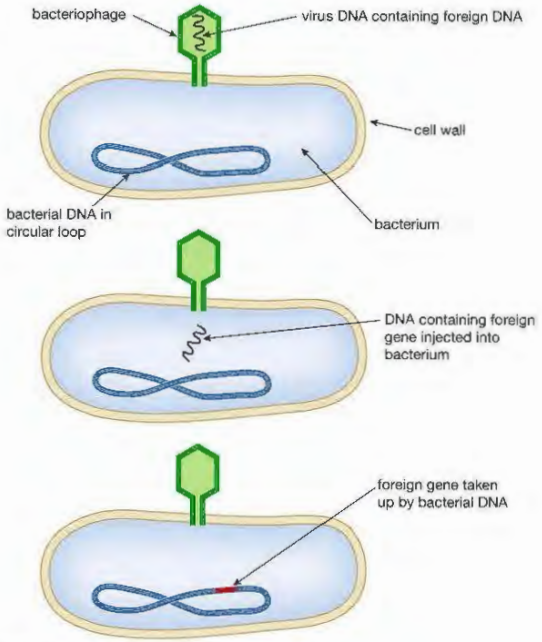
Biologists now had a method of transferring a gene from any cell into a bacterium. They could insert the gene into a plasmid and then transfer the plasmid into a bacterium. The plasmid is called a **vector** because it is the means of transferring the gene. The main processes involved in producing a transgenic bacterium are shown in Figure 22.4.



▲ Figure 22.4 Stages in producing a transgenic bacterium.

Another vector that has been used to introduce foreign DNA into bacterial cells is the **bacteriophage**. A bacteriophage, or 'phage', is a virus that attacks a bacterium. It does this by attaching to the cell wall of the bacterium and injecting its own DNA into the bacterial cell (Figure 22.5). This DNA becomes incorporated into the DNA of the host cell, and eventually causes the production of many virus particles.

If a foreign gene can be inserted into the DNA of the virus, the virus will inject it into the bacterium along with its own genes. Viruses were used as vectors in the early days of genetic modification, but most gene transfer is now carried out using plasmids.



▲ Figure 22.5 A bacteriophage attacking a bacterial cell.

MAKING USE OF GENETICALLY MODIFIED BACTERIA

Different bacteria have been genetically modified to manufacture a range of products. Once they have been genetically modified, they are cultured in fermenters to produce large amounts of the product (see Chapter 21). Some examples are described below.

DID YOU KNOW?
More insulin is required every year because the number of diabetics increases worldwide each year and diabetics now have longer life spans.

1. **Human insulin** People suffering from diabetes need a reliable source of insulin. Before the use of genetic engineering to make human insulin, the only insulin available was extracted from the pancreases of other animals such as cattle or pigs. This is not quite the same as human insulin and does not give the same level of control of blood glucose levels.
2. **Enzymes for washing powders** Many stains on clothing are biological. Blood stains are largely proteins, grease marks are largely lipids. Enzymes can digest these large, insoluble molecules into smaller, soluble ones. These then dissolve in the water. Amylases digest starch, proteases digest proteins and lipases digest lipids. Bacteria have been genetically engineered to produce enzymes that work at higher temperatures, allowing even faster and more effective action.

3. **Enzymes in the food industry** One bacterial enzyme used in the food industry is glucose isomerase. This enzyme catalyses a reaction which converts glucose into a similar sugar called fructose. Fructose is much sweeter than glucose and so less is needed to sweeten foods. This has two advantages – it saves money (since less is used) and it means that the food contains less sugar and is healthier.

DID YOU KNOW?
Before human growth hormone from genetically modified bacteria was available, the only source of the hormone was from human corpses. This was a rather gruesome procedure and had health risks. A number of children treated in this way developed Creutzfeld–Jacob disease (the human form of 'mad cow' disease). When this became apparent, the treatment was withdrawn.

4. **Human growth hormone** The pituitary gland of some children does not produce enough of this hormone and they show a slow rate of growth. Injections of growth hormone from genetically modified bacteria restore normal growth patterns.
5. **Bovine somatotrophin** *(BST)* (a growth hormone in cattle) This hormone increases the milk yield of cows and increases the muscle (meat) production of bulls. Giving injections of BST to dairy cattle can increase the milk yield by up to 10 kg per day. To do this they need more food, but this increased cost is more than offset by the increased income from the increased milk yield (Table 22.1).
6. **Human vaccines** Bacteria have been genetically modified to produce the antigens of the hepatitis B virus. This is used in the vaccine against hepatitis B. The body makes antibodies against the antigens but there is no risk of contracting the actual disease from the vaccination.

Table 22.1 Effects of BST on milk yield.

	Feed / kg per day	Milk output / kg per day	Milk to feed ratio
without BST	34.1	27.9	0.82
with BST	37.8	37.3	0.99

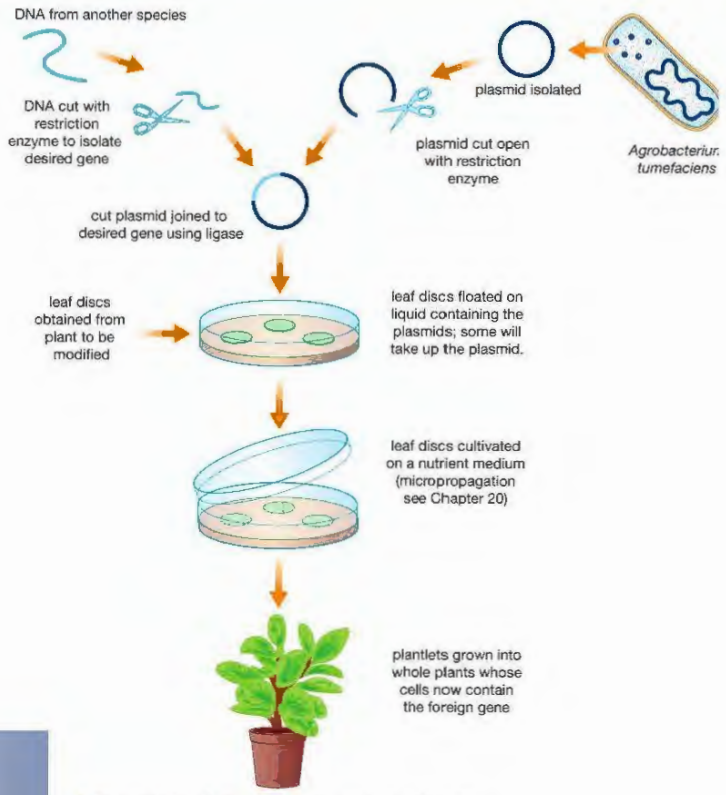
Since the basic techniques of transferring genes were developed, many unicellular organisms have been genetically modified to produce useful products. Techniques for transferring genes into plants and animals have also been developed.

PRODUCING GENETICALLY MODIFIED PLANTS

The gene technology described so far can transfer DNA from one cell to another cell. In the case of bacteria, this is fine – a bacterium only has one cell. But plants have billions of cells and to genetically modify a plant, each cell must receive the new gene. So, any procedure for genetically modifying plants has two main stages:

- introducing the new gene or genes into plant cells
- producing whole plants from just a few cells.

Biologists initially had problems in inserting genes into plant cells. They then discovered a soil bacterium called *Agrobacterium*, which regularly inserts plasmids into plant cells. Now that a vector had been found, the rest became possible. Figure 22.6 outlines one procedure that uses *Agrobacterium* as a vector.



▲ Figure 22.6 Genetically modifying plants using *Agrobacterium*.



▲ Figure 22.7 A gene gun

This technique cannot be used on all plants. *Agrobacterium* will not infect cereals and so another technique was needed for these. The 'gene gun' was invented. This is, quite literally, a gun that fires a golden bullet (Figure 22.7). Tiny pellets of gold are coated with DNA that contains the desired gene. These are then 'fired' directly into plant tissue. Research has shown that if young, delicate tissue is used, there is a good uptake of the DNA. The genetically modified tissue can then be grown into new plants using the same micropropagation techniques as those used in the *Agrobacterium* procedure. The gene gun has made it possible to genetically modify many cereal plants as well as tobacco, carrot, soybean, apple, oilseed rape, cotton and many others. There is, however, much debate about the rights and wrongs of GM crops. You may like to carry out some further research on this topic.

MAKING USE OF GENETICALLY MODIFIED PLANTS

Large numbers of genetically modified plants are already available to plant growers and farmers.

Some fruit and vegetables have been engineered to have extended 'shelf lives' – they last longer before they start to go bad. Other crop plants have been modified to be resistant to herbicides (weedkillers). This allows farmers to spray herbicides at times when they will have maximum effect on the weeds, without affecting the crop plant.. There are concerns that this will encourage farmers to be less careful in their use of herbicides. In another example, genes from Arctic fish that code for an 'anti-freeze' in their blood have been transferred to some plants to make them frost resistant.

The gene gun allowed biologists to produce genetically modified rice called 'golden rice' (Figure 22.8). This rice has three genes added to its normal DNA content. Two of these come from daffodils and one from a bacterium. Together, these genes allow the rice to make beta-carotene – the chemical that gives carrots their colour. It also colours the rice, hence the name 'golden rice'. More importantly, the beta-carotene is converted to vitamin A when eaten. This could save the eyesight of millions of children in less economically developed countries, who go blind because they do not have enough vitamin A in their diet.

EXTENSION WORK

Golden rice sounds like a good idea, but there have been several problems with it. Some people believe that there are ethical and environmental reasons why golden rice should not be grown and that it is better to provide other, natural crops containing enough beta-carotene. You could carry out some research into the advantages and disadvantages of golden rice.



▲ Figure 22.8 Golden rice

KEY POINT

What if you could receive a 'vaccination' every time you ate a banana? Scientists are researching the possibility of transferring the genes that produce the antigens for certain diseases to bananas. If they succeed, then when you eat the banana, the antigens will stimulate an immune response. There will be no risk of you catching the disease – and no need to have needles stuck in you!

Genetically modified plants are also helping humans to resist infection. Biologists have succeeded in modifying tobacco plants and soybeans to produce antibodies against a range of infectious diseases. If these can be produced on a large scale, they could be given to people who are failing to produce their own antibodies. Other modified tobacco plants produce the hepatitis B antigens that could be used as the basis for a vaccine. There is always a risk with a vaccine containing viruses that they may somehow become infectious again. This could not happen with a vaccine containing only plant-produced antigens.

Besides the specific examples given, research into the genetic modification of plants provides, or hopes to provide, plants with:

- increased resistance to a range of pests and pathogens
- increased heat and drought tolerance
- increased salt tolerance
- a better balance of proteins, carbohydrates, lipids, vitamins and minerals.

In addition, some genetically modified oilseed rape plants will be used in large-scale production of biodegradable plastics and anti-coagulants.

One of the biggest achievements would be to modify crop plants like cereals and potatoes to allow nodules of nitrogen-fixing bacteria to form on their roots. At the moment only legumes (peas, beans and other plants with seeds in 'pods') can do this. Biologists know that the ability is genetically controlled. So far, they have not been able to transfer these genes to other plants. If they could, vast areas of infertile soil would be able to yield good crops of cereals without the need to use large quantities of fertilisers.

The bacteria in the root nodules would obtain nitrogen from the air in the soil and 'fix' it in a more usable form (usually ammonia). By doing this, they would make a supply of usable nitrogen available to the plants. The plants would convert this into plant protein and use the protein for growth. The cost of producing these crops would decrease dramatically.

PRODUCING GENETICALLY MODIFIED ANIMALS

Producing genetically modified animals presents some of the same problems as those connected with modifying plants. Animals, like plants, are multicellular. It is not enough simply to transfer a gene to a cell. That cell must then grow into a whole organism. The plasmid technology used to create genetically modified plants depends on the modified cells being grown into whole plants using micropropagation. No such micropropagation techniques exist for animals.

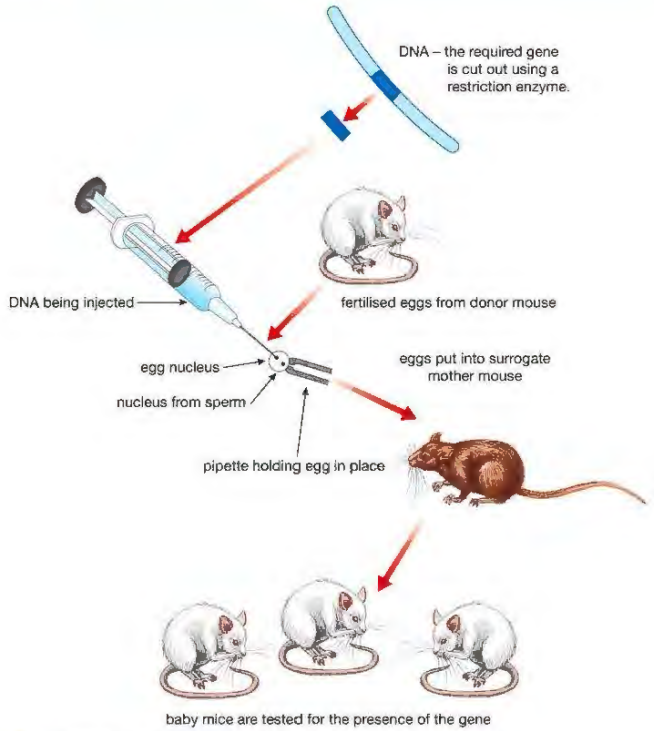
Scientists researching the production of genetically modified animals had to find other techniques. The most successful involves injecting DNA directly into a newly fertilised egg cell. This develops into an embryo, then an adult (Figure 22.9).

Research of this kind can produce beneficial results similar to those achieved by genetically modifying plants, such as:

- manufacture of human proteins, such as antibodies, blood clotting factors or alpha-1-antitrypsin (AAT) – see below
- increased production of a particular product, e.g. higher milk yield in cows, greater muscle mass in animals used for meat
- increased resistance to disease
- production of organs for transplantation (xenotransplantation).

DID YOU KNOW?

'Xenotransplantation' means transplanting organs from other animals into humans. Transgenic pigs have been produced with genes that code for the main human 'marker antigens'. The cells of the pig's organs therefore have these human antigens on their surface and the organs would be less likely to be rejected by a recipient. If this became possible on a large scale, it could help to overcome the shortage of donor organs for transplantation. However, many people have ethical concerns about using animal organs for transplanting into humans.



▲ Figure 22.9 The procedures used in producing genetically modified animals.

BIOLOGY ONLY

CLONING TRANSGENIC ANIMALS

There is also the potential to **clone** genetically modified animals (see Chapter 20). This would make it possible to produce large numbers of animals, all genetically identical. The cloned animals could then be used for production of human proteins.

For example, the human protein alpha-1-antitrypsin (AAT) is involved in the immune response. When phagocytes fight an infection, they produce the enzyme trypsin. AAT is produced as a counter measure to stop trypsin attacking normal tissues.

KEY POINT

In the lungs, a deficiency of AAT causes a rare type of emphysema with symptoms similar to the emphysema caused by smoking – see Chapter 3.

Some people carry a mutation that means they do make enough AAT. The trypsin then damages normal tissues, mainly in the liver and lungs.

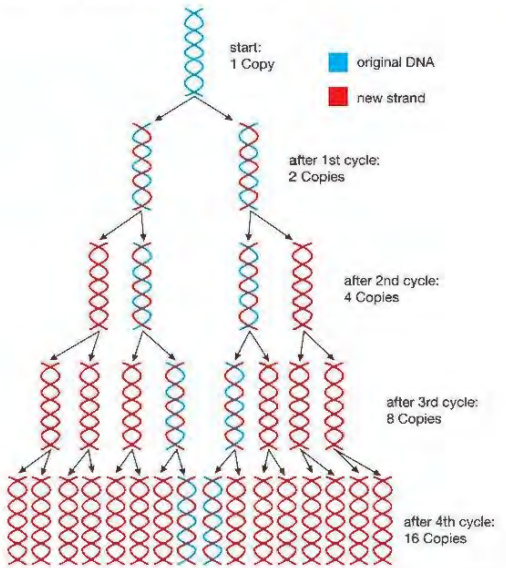
Transgenic sheep have been genetically modified to produce AAT in their milk. It may be possible to use this to treat people with AAT deficiency. If the sheep were to be cloned, this would increase the amount of AAT available.

END OF BIOLOGY ONLY

LOOKING AHEAD – THE POLYMERASE CHAIN REACTION

The polymerase chain reaction (PCR) is a laboratory process that amplifies DNA. This means that it rapidly produces a large number of copies of a piece of DNA, starting from as little as one molecule. The invention of PCR revolutionised gene technology and is used in virtually every application of the science.

Each cycle of PCR doubles the amount of DNA, producing an exponential increase in the numbers of copies (Figure 22.10). Thirty cycles produces over a billion copies of the original DNA.



▲ Figure 22.10 PCR produces an exponential increase in the number of copies of a piece of DNA.

The procedure uses short lengths of single-stranded DNA called primers, which are precisely complementary to one end of each strand of the DNA you wish to amplify. The DNA is heated to separate the strands and the primers added to the ends of each strand. An enzyme called DNA polymerase builds up two new complimentary strands by adding nucleotides - one at a time - along each strand.

Nowadays, PCR is an automated process and can be performed in as little as 2–3 hours in a PCR machine. You just place small tubes containing the DNA sample, primers, free nucleotides, DNA polymerase and a buffer solution in the machine, switch on, and wait for the results.

CHAPTER QUESTIONS

More questions on gene technology can be found at the end of Unit 6 on page 301.

SKILLS CRITICAL THINKING

1 Why is it relatively easy to genetically modify bacteria?

A they reproduce slowly and accurately
B they are able to take up plasmids
C they only contain a single chromosome
D they make restriction enzymes

2 The statements below show some stages in the production of human insulin from genetically modified bacteria.

1. DNA for insulin inserted into plasmids
2. bacteria cloned
3. plasmids inserted into bacteria
4. DNA for insulin cut out using restriction enzyme

Which of the following shows the correct sequence of steps in the process?

A 2 → 1 → 4 → 3
B 4 → 2 → 3 → 1
C 4 → 1 → 3 → 2
D 2 → 3 → 4 → 1

3 Which of the following genetically modified plants have not been produced yet?

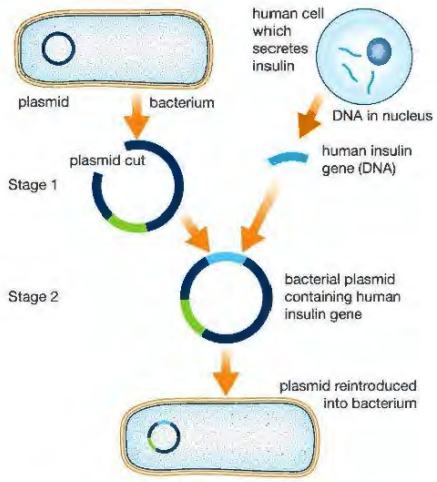
A rice that is rich in beta-carotene to make vitamin A
B plants resistant to weedkiller
C plants containing an anti-freeze
D cereals with nitrogen-fixing bacteria in root nodules

4 Which of the following enzymes is used to join together pieces of DNA?

A ligase
B DNA polymerase
C protease
D restriction enzyme

SKILLS ANALYSIS 8

5 The diagram shows the main stages in transferring the human insulin gene to a bacterium.



a Name the enzymes used at stages 1 and 2.

SKILLS CRITICAL THINKING 8

b What is the role of the plasmid in this procedure?

c How would the insulin-producing bacteria be used to produce significant amounts of insulin?

SKILLS REASONING 9

d Why is the insulin produced this way preferred to insulin extracted from other animals?

SKILLS INTRAPERSONAL 12

6 Carry out some research to find out more about the use of transgenic animals for organ transplantation. What are the arguments for and against its use? What are the ethical objections?

SKILLS CRITICAL THINKING 8

7 Producing genetically modified plants and animals is more complex than producing genetically modified bacteria.

a Describe two ways in which genes can be introduced into plant cells.

b How are these genetically modified cells used to produce whole organisms?

c What sort of animal cell is genetically modified and then used to produce the whole organism?

SKILLS INTRAPERSONAL 12

8 Write an essay about the importance of genetic engineering. In your essay you should make reference to:

a important potential benefits resulting from genetic engineering in animals and plants

b concerns about the risks resulting from genetic engineering in animals and plants.

UNIT QUESTIONS

SKILLS CRITICAL THINKING 8 **1**

Some experimental sheep were genetically modified, using a human gene. The gene codes for a protein involved in blood clotting, called factor IX.

The gene for factor IX was cut from human DNA and injected into a sheep body cell. The nucleus of this body cell was transferred to an enucleated sheep ovum. The ovum was cultured in a dish and then implanted in the uterus of a surrogate sheep to complete its development.

Two lambs produced by this method grew to maturity. They produced milk from which Factor IX could be extracted.

a Name an enzyme that can be used to cut DNA. **(1)**

7 8

b What is an *enucleated ovum*? **(2)**

c The sheep produced by this method are called *transgenic*. What is meant by this term? **(1)**

SKILLS REASONING 7

d The transgenic sheep were reproduced by *cloning*. What are the advantages of producing the sheep in this way? **(3)**

6

e Factor IX can be used to treat people with a blood clotting disorder called haemophilia. Why is important to have blood that clots? **(2)**

SKILLS CRITICAL THINKING 5

f Blood contains plasma, white blood cells, red blood cells and platelets. Which *two* components of the blood are involved with blood clotting? **(2)**

Total 11 marks

SKILLS CRITICAL THINKING 9 **2**

Some genes are transferred from one plant species to another. These genes are called 'jumping genes'. Environmentalists are concerned that genetically modified plants may transfer some of their genes to wild species. Explain their concern about genetically modified plants that:

a have genes that make them resistant to herbicides (weedkillers) **(2)**

b have genes that make them resistant to pests **(2)**

c have genes that increase the yield of the crop they can produce. **(2)**

Total 6 marks